Flexibility Method Accelerated by H-matrices

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Flexibility method accelerated by $\mathcal{H}$-matrices

Pressure p-0, displacement p-1

Element formulation

To be more aligned with the BEM and potentially with the LBB, we can assign pressure degrees of freedom p_i (or tractions $\mathbf{t}_i$) to element centers and displacement degrees of freedom $\mathbf{u}_j$ to nodes. Then the number of pressure DOFs will be defined by the number of faces and the type of problem $m = d_p \cdot N_\mathrm{e}$, where $d_p = 1$ for frictionless problems and $d_p = \dim$ for frictional problems. The number of displacement DOFs is given by the dimension and the number of nodes on the surface of interest $n = \dim \cdot N_n$.

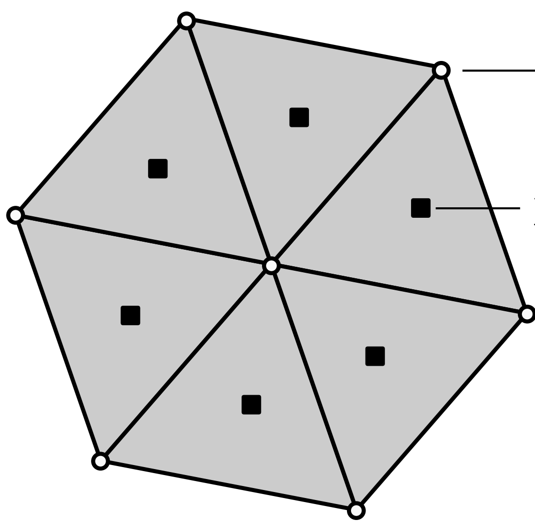

Figure 1: Elements and degrees of freedom

Matrix construction

Within this formulation the flexibility matrix $S_c \in \mathbb{R}^{n\times m}$ is not square. Then the following relation is consistent:

$$[g]_n = [S_c]_{n\times m}[p]_m + [g_0]_n,$$

where $[p]$ is the array of pressure DOFs, and $[g]$, $[g_0]$ are the arrays of gap and initial gaps defined at nodes.

To construct the matrix S_c, we use, for example, the direct sampling method, which consists in applying pressure p_0 at individual elements one-by-one and by measuring induced displacement. It is done by solving FE problem:

$$[K][u]^i = [f]^i,$$

where $[f]^i$ represents the right hand side obtained through integration

$$l(\mathbf{v}) = \int\limits_{\mathrm{el}_i} p_0 \mathbf{n}\cdot\mathbf{v}\, dS,$$

as shown in the figure below.

The S_c's matrix i-th row is then given by

$$S_{c_{i,\cdot}} = \frac{1}{p_0}[u]^i.$$

In practice, PETSc solver allows to assemble all required $[f]^i$ in a single matrix $[F]_{N_{\mathrm{DOF}}\times m}$, where N_{DOF} is the total number of DOFs of the FE mesh. Then, by solving the following system in batch mode:

$$[K]_{N_{\mathrm{DOF}}\times N_{\mathrm{DOF}}}[X]_{N_{\mathrm{DOF}}\times m} = \frac{1}{p_0}[F]_{N_{\mathrm{DOF}}\times m},$$

we obtain directly a matrix $[X]$ whose entries at surface nodes can be easily extracted to obtain $[S_c]_{n\times m}$ matrix.

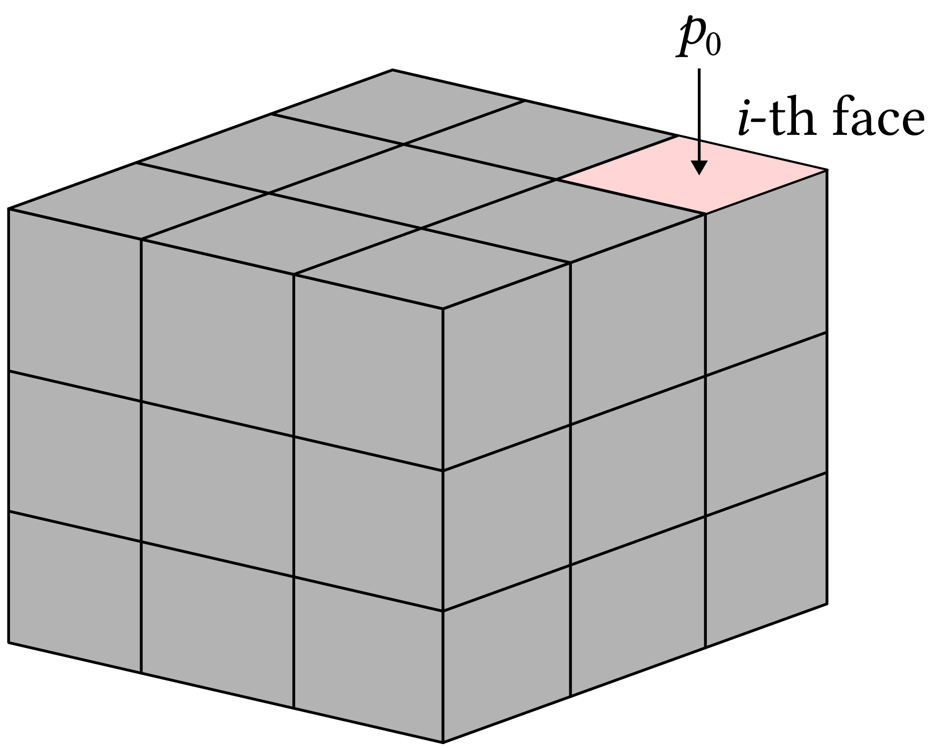

Figure 2: Direct sampling matrix construction